Assignment A2 - due 16/9

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Part 1 - Basic description of language development

1. **Describe your sample (n, age, gender, clinical and cognitive features of the two groups) and critically assess whether the groups (ASD and TD) are balanced**

Descriptive statistics for the overall sample

The groups are relatively balanced according to ethnicity (the study was executed in Denmark), non-verbal IQ and number of children in each group. The participants were, however, mostly male.
Overall assessment of the data:

- n = 372 (ASD = 176, TD = 196)
- men > women in both groups (ADS: 150 M, 26 F, TD: 160 M, 36 F)
- ethnicity = white > other ethnicities (ASD: 130 white kids, other minorities: 46, TD: 190 white kids, other minorities: 6)
- Non-verbal IQ: ASD mean 33.15, TD mean 35.8

We see clear differences between ASD and TD when it comes to measures of verbal IQ and ADOS. Across all visits we also see the following difference in their linguistic capacities.

- We see a clear difference in autistic measure, ADOS (mean of ADS = 13.8, mean of TD = 1.2)
- Verbal IQ: ASD mean 22.62, TD mean: 29.29
- Mean MLU for the children: ASD: 1.6, TD: 2.3
- Mean Types (unique words used): ASD: 77, TD: 129
- Mean Tokens (amount of words used): ASD: 293, TD: 475

Descriptive statistics for visit 1

The ASD children were matched with TD children by their linguistic skills. This can be seen by the descriptive statistics of the children's linguistic skills at their first visit. This entails that the age difference between the two groups is approximately 1 year at their first visit.

- Age (in months): mean at first visit ASD: 33, TD: 20.4
- The children had similar mean MLU at first visit: TD: 1.37, ASD: 1.24
- Mean Types at first visit (unique words used): ASD: 42, TD: 45
- Mean Tokens at first visit (amount of words used): ASD: 161, TD: 194

2. **Describe linguistic development (in terms of MLU over time) in TD and ASD children (as a function of group).**

Motivation for determining predictors

Fixed effects: As a measure of time, we have chosen to enter “age” and not “visit” in our model. Taken visit by visit, there is a skew in age since autistic children had a higher age at each visit than typically developing children. Thus, “visit” has a skewed representation of the children’s age. Furthermore, the time between each visit differs from visit to visit. Using ‘age’ as a predictor and not ‘visit’ accounts for this difference. However the limitation of entering age is that there is a difference in the maximal age for the two groups. TD have a max age of 45 months, whereas ASD have a max age of 62 months.

Random effects: We had random intercepts for each subject and random slopes for age to account for the subjects’ individual differences in both their skills at start and their different learning speed.

Eyeballing the data

Eyeballing the data gave the indication that there might be an interaction effect between age and diagnosis, and that the random effects are important especially for the ASD group where participants have very different developments (see figure 1).

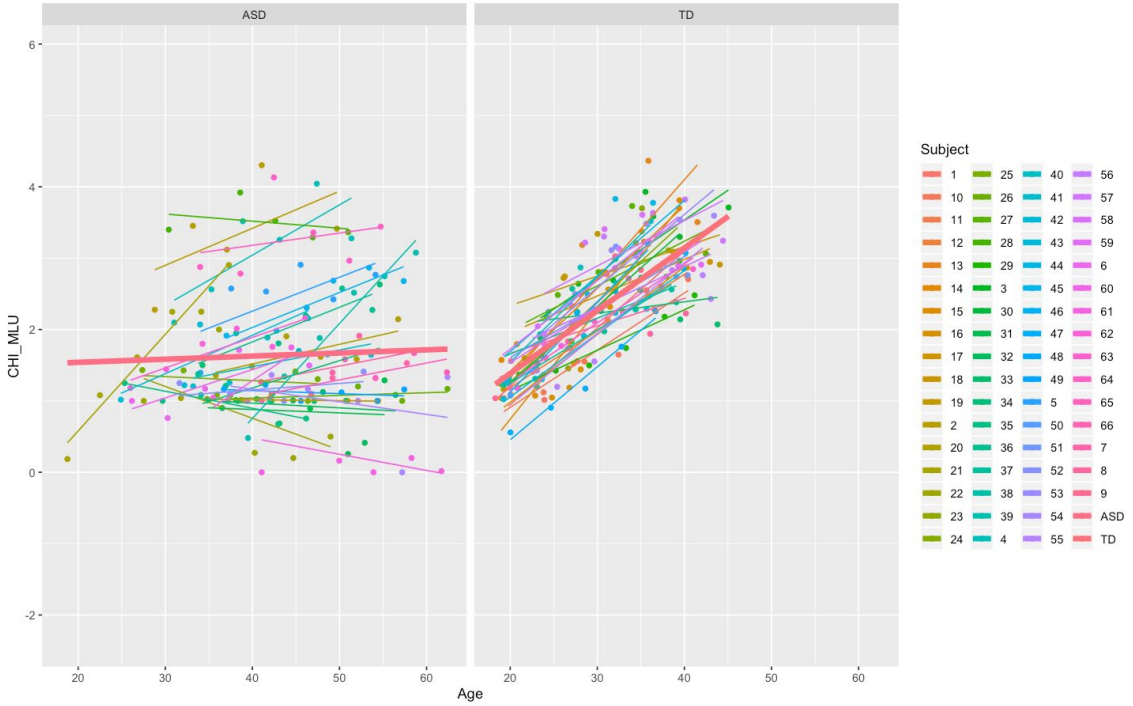

Figure 1: CHI_MLU plotted against age. Random slopes and intercept for each participant.

The model

We have built a linear mixed effects growth model using the following syntax and package lme4:

*model = lmer(child_MLU ~ Diagnosis * Age + (Age|Subject) + I(Age^2))*

Results

There was a significant interaction effect between diagnosis and age, $\beta = 0.030$, $SE = 0.012$, $t = 2.542$, $p < .01$.
The interaction effect makes the main effects of diagnosis and age, respectively, useless, since they can only be interpreted when taking the other into account.
Fixed effects alone accounted for roughly 38 % of the variance. Fixed effects and random effects together accounted for roughly 82 % of the variance.
An anova comparing our model with a model excluding the interaction Diagnosis:Age, showed that the interaction made our model significantly better (BIC = 681 and BIC = 612 respectively), thus confirming the importance of the interaction.

Description of results

Children who are typically developing have a significantly higher mean length of utterance as they develop compared to children who have an autistic spectrum disorder. For each month in their development the mean length/utterance of typical children are 0.003 syllables longer than children with autism spectrum disorder. This is the interaction effect.

Plot of the models predictions

Figure 2 shows the actual values (blue) plotted against the predicted values (red).
However these predictions are not sensible, since we didn't split the dataset into training and testing. The predicted values merely show that our model is good at explaining the data.

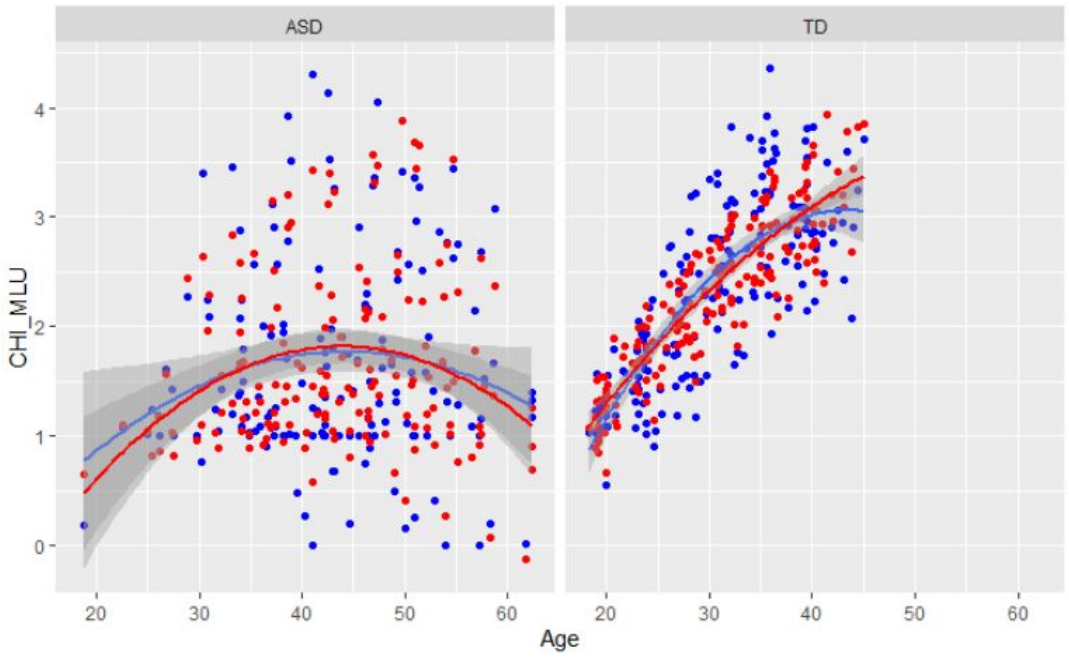

Figure 2: actual values of the mean length of utterance for children (blue) compared to fitted values (red).

3. **Describe how parental use of language (in terms of MLU) changes over time. What do you think is going on?**

Eyeballing the data

Eyeballing the data showed that an interaction effect between age and diagnosis probably was not present. It also showed that random effects would not account for as much variance as it did with child mean length utterances.

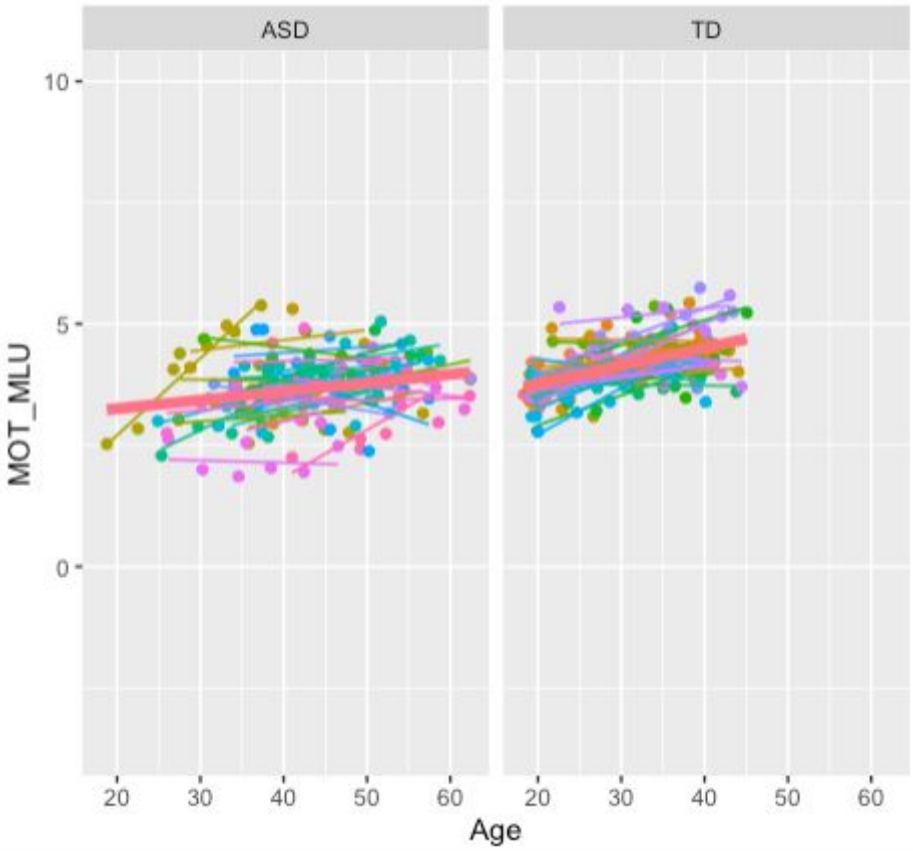

Figure 3: Mother MLU according to the diagnosis and age of the child

Results

To assess the parental use of language (also in terms of MLU), a model was built using the following syntax:

model = lmer(Mother_MLU ~ Diagnosis + Age + (Age|SUBJ), df)

There was a significant effect of diagnosis; the mother of the child had a lower MLU if the child had the diagnosis of ASD, $\beta = 2.53$, $SE = 0.18$, $t(100.1) = 14.375$, $p < .0001$, compared to the MLU of mothers with typically developing children, $\beta = 3.314$, $SE = 0.12$, $t(61.73) = 6.31$, $p < .0001$. Moreover, there was a significant effect of age; the older the child got (in months) the larger MLU of the mother, $\beta = 0.0277$, $SE = 0.004$, $t(58.95) = 6.865$, $p < .0001$.

Fixed effects alone accounted for roughly 20 % of the variance. Fixed effects and random effects together accounted for roughly 68 % of the variance.

It seems that the mothers of typically developing children (TD) talk more to their children than mothers of children with ASD, and that the MLU of mothers increases as the age of the child increases. This makes sense since TD children have a larger MLU than ASD children.

The results are depicted in figure 4.

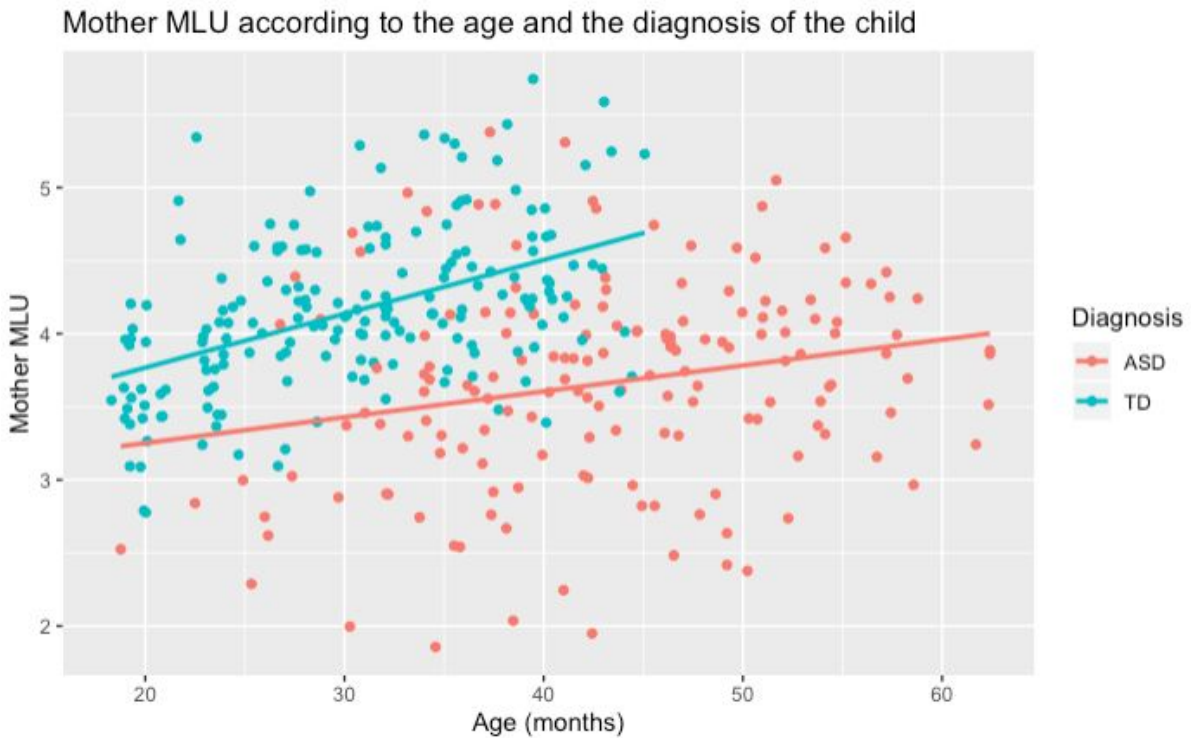

Figure 4: Mother MLU according to the age and diagnosis of the child

4. Include individual differences in your model of language development (in children). Identify the best model.

We have conducted a post-hoc exploratory analysis to see the possible effect of other independent variables.
We found that ‘ethnicity’ was a significant predictor. However, this variable is not very well balanced in each level (e.g. there is only one asian subject), thus, we discard this variable as a valid predictor.
‘Gender’ was found to be non-significant.

The variables ‘non-verbal-IQ’, ‘verbal-IQ’ and ‘Socialization’ were all found to be significant. These results can be used to make new hypotheses in a follow up study. However, since we did not include them in our hypothesis we can’t use them in the current analysis.